Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A transformer with efficiency 80% works at 4 kW and 100 V. If the secondary voltage is 200 V, then the primary and secondary currents are respectively

A. 40 A and 16 A

B. 16 A and 40 A

C. 20 A and 90 A

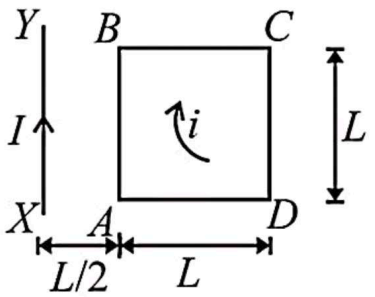
D. 40 A and 20 A

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A square loop ABCD carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be:



A. $\frac{2\mu_0 Ii}{3\pi}$

B. $\frac{\mu_0 Ii}{2\pi}$

C. $\frac{2\mu_0 IiL}{3\pi}$

D. $\frac{\mu_0 IiL}{2\pi}$

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The coefficient of mutual inductance, when magnetic flux changes by 2×10^{-2} Wb and current changes by 0.01 A is

A. 2H

B. 4H

C. 3H

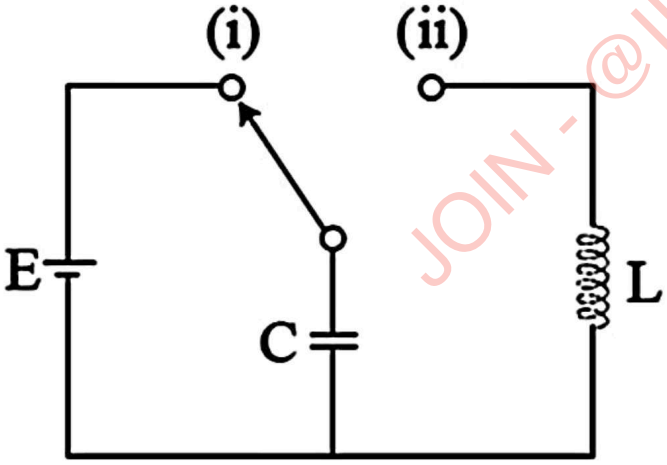
D. 8H.

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the following electrical network at $t < 0$, key is placed on (i) till the capacitor got fully charged. Key is placed on (ii) at $t = 0$. Time when the energy in both the capacitor and the inductor will be same for the first time is:



A. $\frac{\pi}{4}\sqrt{LC}$

B. $\frac{3\pi}{4}\sqrt{LC}$

C. $\frac{\pi}{3}\sqrt{LC}$

D. $\frac{2\pi}{3}\sqrt{LC}$

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A paramagnetic sample shows a net magnetization of 8Am^{-1} when placed in an external magnetic field of $0.6T$ at a temperature of $4K$. When the same sample is placed in an external magnetic field of $0.2T$ at a temperature of $16K$, the magnetization will be :

A. $\frac{32}{3}\ \text{Am}^{-1}$

B. $\frac{2}{3}\ \text{Am}^{-1}$

C. 6Am^{-1}

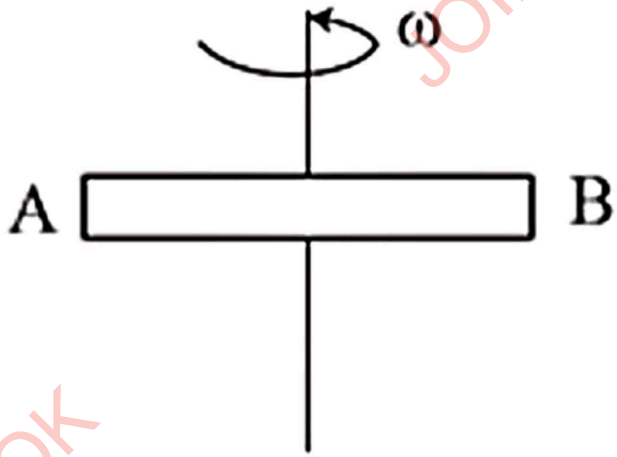
D. 2.4Am^{-1}

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A non-conducting rod AB of length l has total charge q with constant charge and mass density. The rod is rotating about an axis passing through centre of mass with constant angular velocity ω as shown in the figure, the magnetic moment of the rod about the axis is :



A. $\frac{q\omega l^2}{12}$

B. $\frac{q\omega l^2}{3}$

C. $\frac{q\omega l^2}{24}$

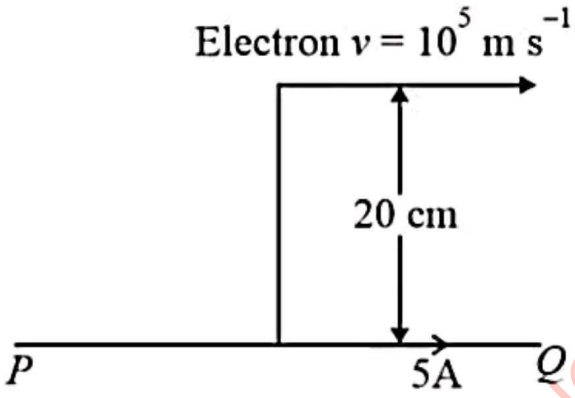
D. $\frac{q\omega l^2}{6}$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An infinitely long straight conductor carries a current of 5A as shown. An electron is moving with a speed of 10^5 m s^{-1} parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.



A. 4×10^{-20} N

B. $8\pi \times 10^{-20}$ N

C. $4\pi \times 10^{-20}$ N

D. 8×10^{-20} N

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements are correct?
(A) Electric monopoles do not exist whereas magnetic monopoles exist.
(B) Magnetic field lines due to a solenoid at its ends and outside can be completely straight and confined.
(C) Magnetic field lines are completely confined within a toroid.
(D) Magnetic field lines inside a bar magnet are not parallel.
(E) $\chi = -1$ is the condition for a perfect diamagnetic material, where χ is its magnetic susceptibility.

Choose the correct answer from the options given below :

A. (C) and (E) only

B. (B) and (D) only

C. (A) and (B) only

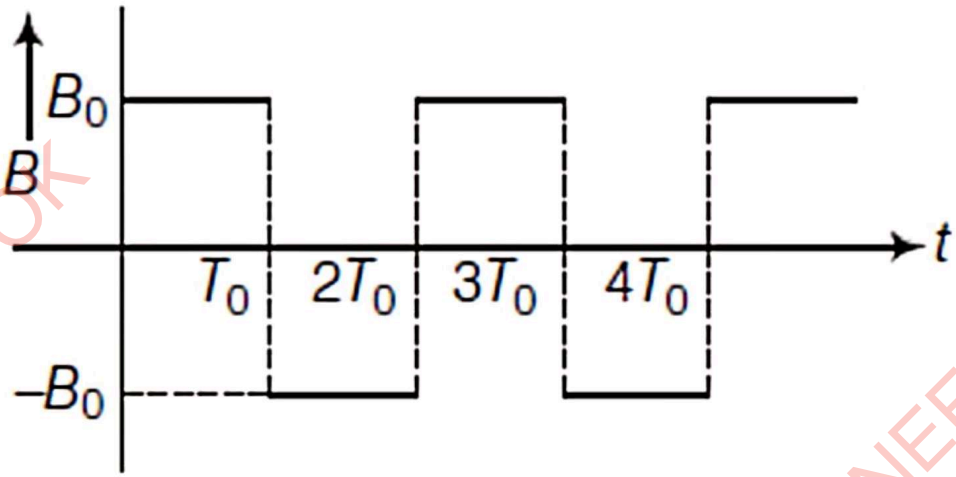
D. (B) and (C) only

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a region, magnetic field along X-axis changes with time according to the given graph. If time period, pitch and radius of helix path are T_0, P_0 and R respectively, then which of the following is incorrect, if the particle is projected at an angle θ_0 with the positive X-axis in $X - Y$ plane?



At $t = \frac{T_0}{2}$, coordinates of charge are

A. $\left(\frac{P_0}{2}, 0, -2R_0\right)$

At $t = \frac{3T_0}{2}$, coordinates of charge are

B. $\left(\frac{3P_0}{2}, 0, 2R_0\right)$

Two extremes from X-axis are at a distance

C. $2R_0$ from each other

Two extremes from X-axis are at a distance

D. $4R_0$ from each other

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The magnetic induction due to a bar magnet of length 6×10^{-2} m and pole strength 5×10^{-3} m at a point 0.1 m away from the centre and along the equator is

A. 3×10^{-9} N A^{-1} m^{-1}, directed from the North Pole to the South Pole.

B. 3×10^{-8} N A^{-1} m^{-1}, directed from the North Pole to the South Pole.

C. 3×10^{-8} T, directed from the South Pole to the North Pole.

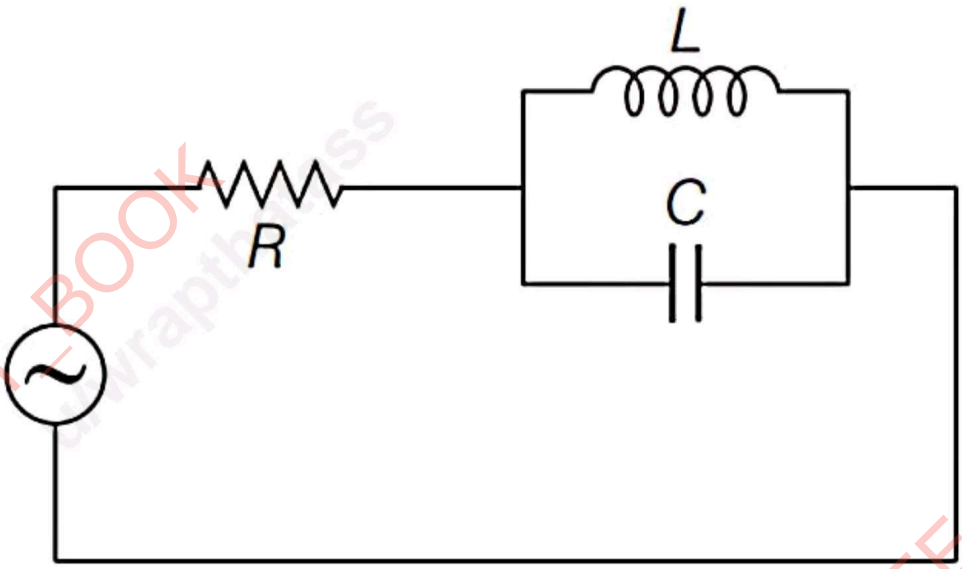
D. 3×10^{-9} T, directed from the South Pole to the North Pole.

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The $L - C - R$ circuit shown below is driven by an ideal AC voltage source.



At frequency $f = \frac{1}{2\pi\sqrt{LC}}$, choose the correct statement.

A. The current through R is zero.

B. The current through R is infinite.

C. The current through R depends on the value of L and C.

D. The current through R depends only on the value of R and not L and C.

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A copper disc of the radius 0.1 m is rotated about its centre with $\omega = 20$ revolutions per second in a uniform magnetic field of 0.1 T with its plane perpendicular to the field. The emf induced across the radius of the disc is

A. $\frac{\pi}{20}$ V

B. $\frac{\pi}{10}$ V

C. 20πmV

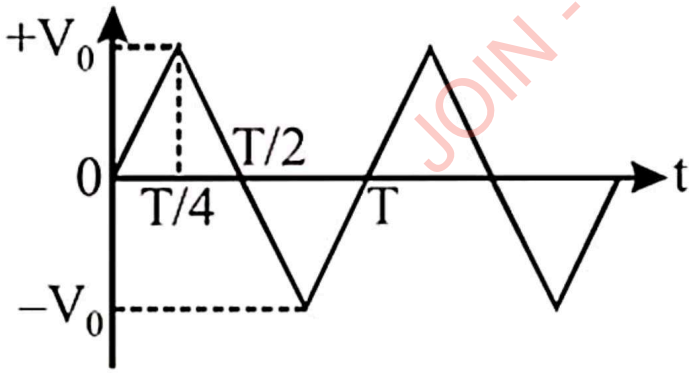
D. 100πmV

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The voltage time (V-t) graph for triangular wave having peak value V_0 is as shown in figure. The rms value of V in time interval from $t = 0$ to $T/4$ is $\frac{V_0}{\sqrt{x}}$ then find the value of x.



A. 5

B. 4

C. 7

D. 3

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A thin rectangular magnet suspended freely has a period of oscillation 4 s. If it is broken into two halves each having half their initial length, then when suspended similarly, the time period of oscillation of each part will be

A. 4 s

B. 2 s

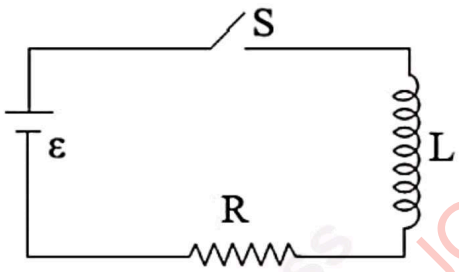
C. 1 s

D. $2\sqrt{2}$ s

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.



Assertion: In the figure, just after closing the switch the potential drop across inductor is maximum.
Reason: The rate of change of current just after closing the switch is maximum.

A. both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B. both Assertion and Reason are true but Reason is not correct explanation of the Assertion.

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Assertion is true but
C. Reason is false.

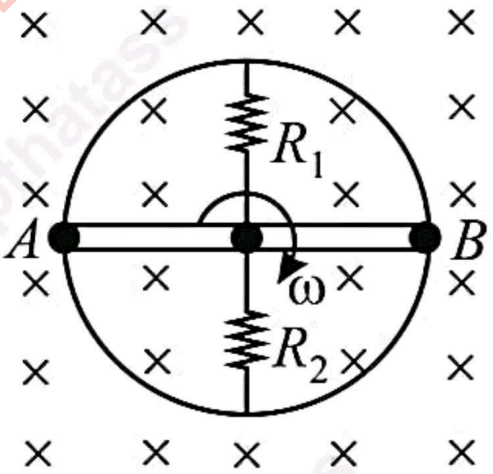
Assertion is false but
D. Reason is true.

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

AB is a resistance less conducting rod which forms a diameter of a conducting ring of radius r rotating in a uniform magnetic field B as shown in the diagram. The resistors R_1 and R_2 do not rotate. Then the current through the resistor R_1 is



A. $\frac{B\omega r^2}{2R_1}$

B. $\frac{B\omega r^2}{2R_2}$

C. $\frac{B\omega r^2}{2R_1R_2}(R_1+R_2)$

D. $\frac{B\omega r^2}{2(R_1+R_2)}$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

There are four light-weight-rod samples A, B, C, D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted

(i) A is feebly repelled

(ii) B is feebly attracted

(iii) C is strongly attracted

(iv) D remains unaffected

Which one of the following is true?

A. C is of a diamagnetic material

B. D is of a ferromagnetic material

C. A is of a non-magnetic material

D. B is of a paramagnetic material

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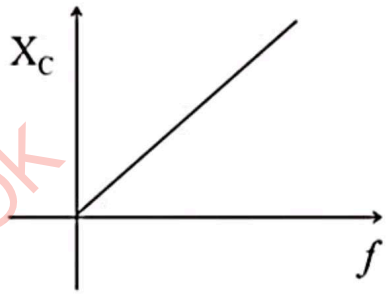
Question No. 19

Only One Option Correct Type

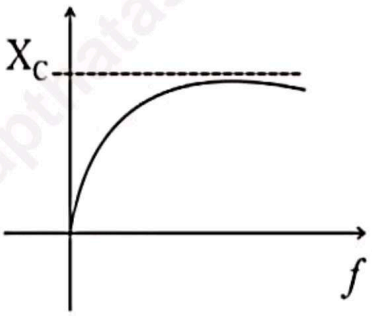
Each question has multiple options out of which ONLY ONE is correct.

Which of the following represents variation of capacitive reactance (X_C) and frequency (f) of the voltage source?

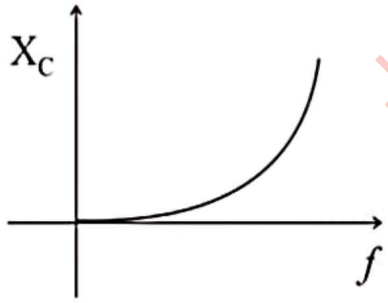
A.



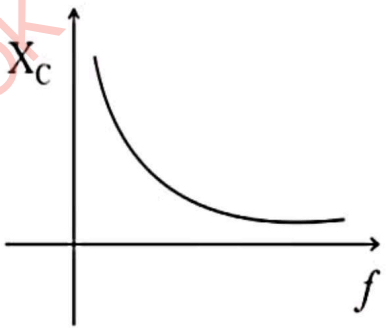
B.



C.



D.



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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In an electrical circuit R, L, C and an AC voltage source are all connected in series. When L is removed from the circuit, the phase difference between the voltage and the current in the circuit is $\pi/3$. If instead, C is removed from the circuit, the phase difference is again $\pi/3$. The power factor of the circuit is

A. $1/2$

B. $1/\sqrt{2}$

C. 1

D. $\sqrt{3}/2$

Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

A long solenoid of the cross-sectional radius R has a thin insulated wire ring tightly put on its winding. One half of the ring has the resistance 10 times that of the other half. The magnetic induction produced by the solenoid varies with time as $B = bt$, whose b is a constant. If the magnitude of the electric field strength in the ring is $\frac{p}{q}\,Rb$, where p and q are the smallest positive integers, then what is the value of $(p + q)$?

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

The coils of a step-down transformer have 500 and 5000 turns. In the primary coil an AC current of 4 A at 2200 V is sent. The value of the current and potential difference in the secondary coil is x A and y V, find value of 3x+y.

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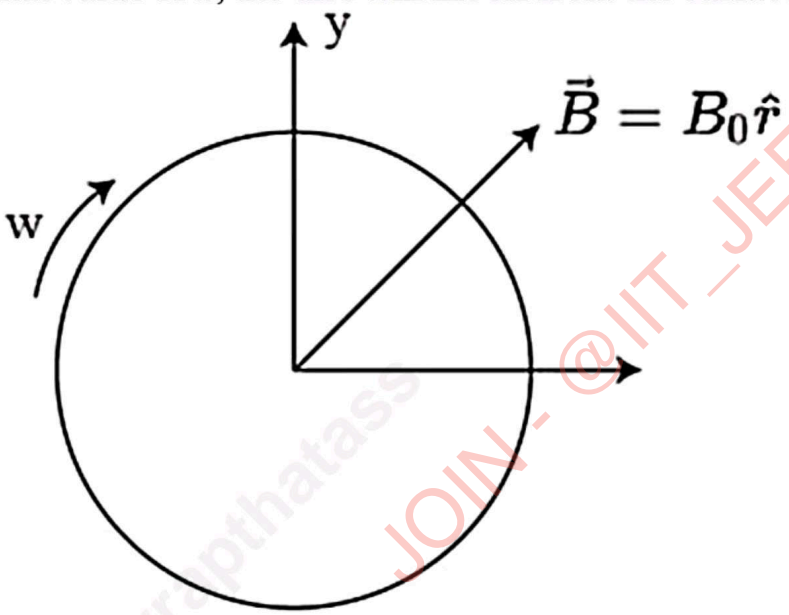
If B is the magnetic field at the centre of a circular loop of area A and its magnetic dipole moment is found to be $n \frac{BA^m}{\mu_0 \sqrt{\pi}}$, then what is the value of $10(\mathrm{n}+\mathrm{m})$?

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A thin non conducting disc of mass $M = 2$ kg, charge $Q = 2 \times 10^{-2}\mathrm{C}$ and radius $R = \frac{1}{6}$ m is placed on a frictionless horizontal plane with its centre at the origin of the coordinate system. A nonuniform, radial magnetic field $\vec{B} = B_0\hat{r}$ exists in space, where $B_0 = 10$ T and $\hat{r}$ is a unit vector in the radially outward direction. The disc is set in motion with an angular velocity $\omega = x \times 10^2 \mathrm{rads}^{-1}$, about an axis passing through its centre and perpendicular to its plane, as shown in the figure. At what value of x, the disc will lift off from the surface.



Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

In a series $L-C-R$ circuit, $C = 10^{-11}$ F, $L = 10^{-5}$H and $R = 100\ \Omega$, when a constant DC voltage E is applied to the circuit, the capacitor acquires a charge 10^{-9}C. The DC source is replaced by a sinusoidal voltage source in which the peak voltage E_0 is equal to the constant DC voltage E. At resonance the peak value of the charge acquired by the capacitor is 10^{-n}C. Mark value of n.

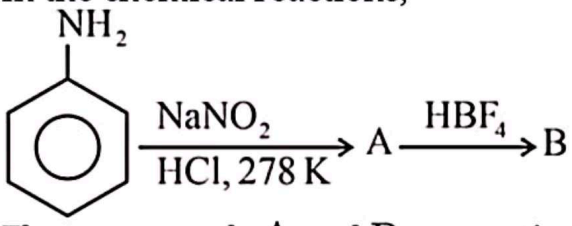
Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the chemical reactions,



The compounds A and B, respectively, are

A. nitrobenzene and fluorobenzene

B. phenol and benzene

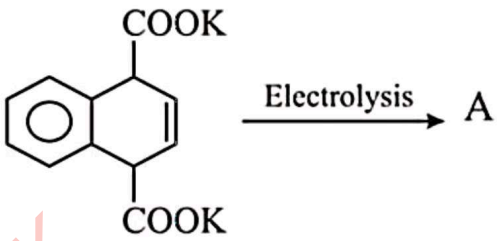
C. benzene diazonium chloride and fluorobenzene

D. nitrobenzene and chlorobenzene

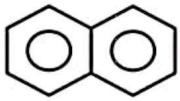
Question No. 2

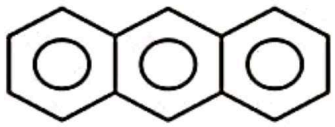
Only One Option Correct Type

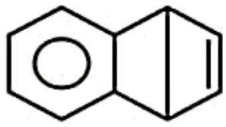
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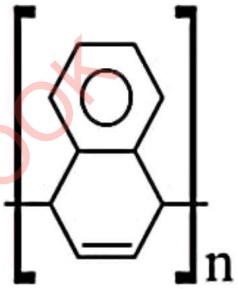


A is :

A. 

B. 

C. 

D. 

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Amine is not formed in the reaction
I. Hydrolysis of RCN
II. Reduction of $\mathrm{RCH=NOH}$
III. Hydrolysis of RNC
IV. Hydrolysis of $\mathrm{RCONH_2}$
The correct answer is

A. I, II and IV

B. I and IV

C. II and III

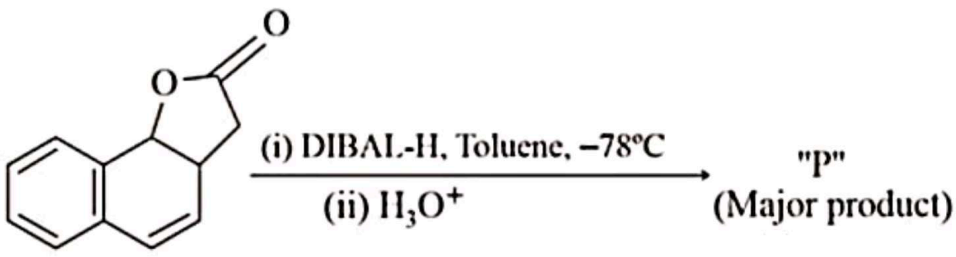
D. I, II and III

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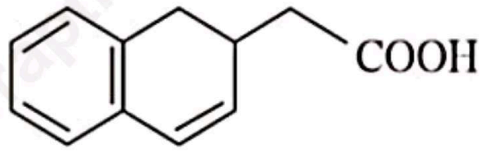
Question No. 4

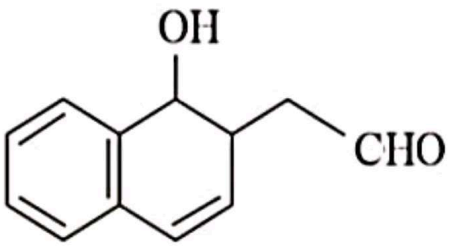
Only One Option Correct Type

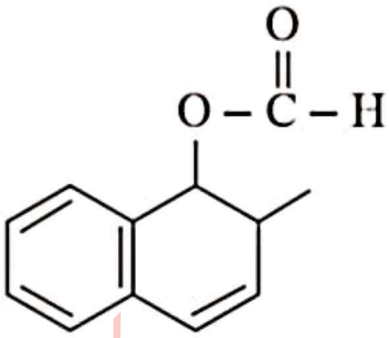
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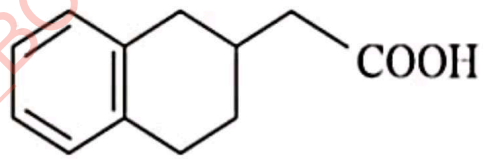


The product "P" in the above reaction is :

A. 

B. 

C. 

D. 

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A: Hinsberg reagent is used for the separation of mixture of 1°, 2° & 3° amines.
R: $C_6H_5SO_2Cl$ does not react with primary amine.

A. both Assertion & Reason are true and the reason is the correct explanation of the assertion.

B. both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

C. Assertion is true statement but Reason is false.

D. both Assertion and Reason are false statements, then mark (4).

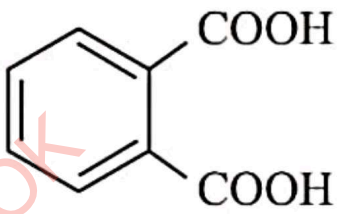
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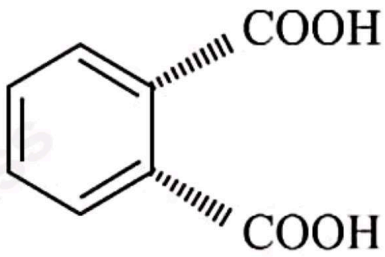
Question No. 6

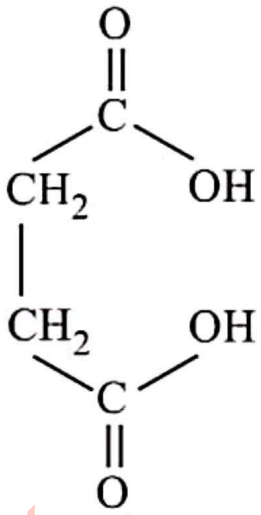
Only One Option Correct Type

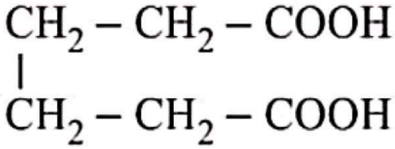
Each question has multiple options out of which ONLY ONE is correct.

With dehydrating agent present which dicarboxylic acid is least reactive towards forming an anhydride?

A. 

B. 

C. 

D. 

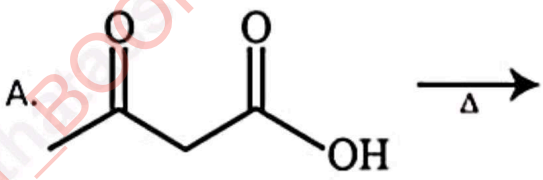
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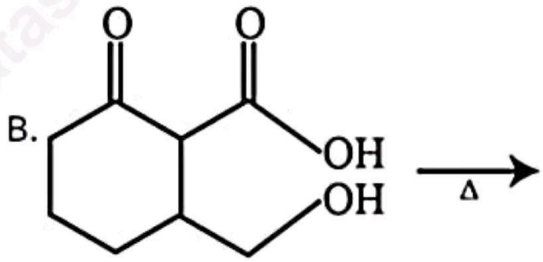
Question No. 7

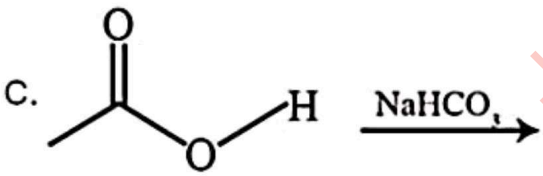
Only One Option Correct Type

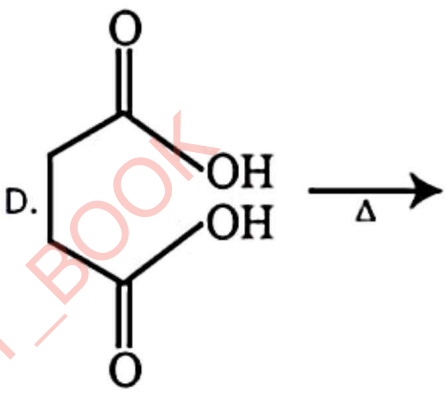
Each question has multiple options out of which ONLY ONE is correct.

In which of the following reaction CO_2 (carbon dioxide) is not released?

A. 

B. 

C. 

D. 

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Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which among the following is a tribasic acid?

A. Oxalic acid

B. Tartaric acid

C. Lactic acid

D. Citric acid

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An amine on treatment with HNO_2 evolved N_2. The amine on exhustive methylation with CH_3I formed a quaternary salt containing 59.07% iodine. The amine is likely to be :

A. CH_3NH_2

B. $(CH_3)_2NH$

C. $C_2H_5NH_2$

D. $(CH_3)_3N$

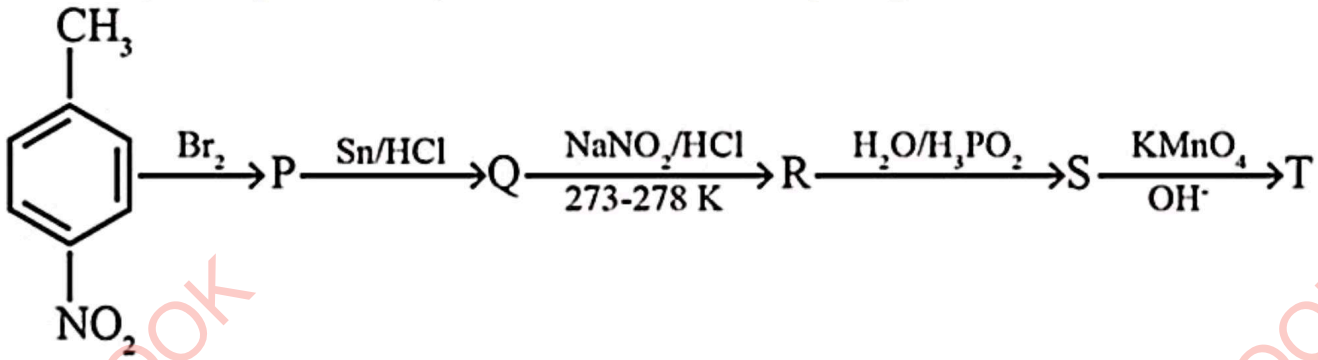
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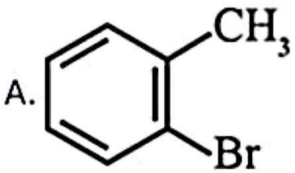
Question No. 10

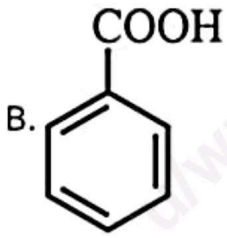
Only One Option Correct Type

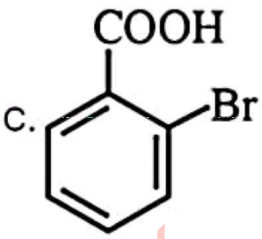
Each question has multiple options out of which ONLY ONE is correct.

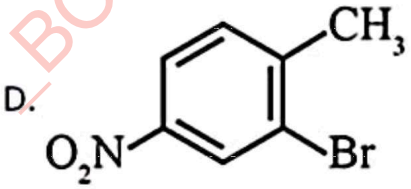
Identify the product (T) in the following sequence of reactions.



A. 

B. 

C. 

D. 

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following will not produce the benzyl amine?

A. Benzamide $\xrightarrow{LiAlH_4}$

B. Phenyl nitrile $\xrightarrow{LiAlH_4}$

C. 2-phenylethyl isonitrile $\xrightarrow{LiAlH_4}$

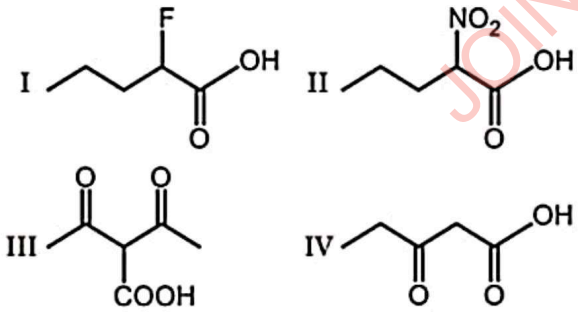
D. 2-phenyl ethanamide $\xrightarrow{Br_2+KOH}$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Arrange the following carboxylic acids in increasing order of reactivity towards thermal decarboxylation reaction



A. $I < II < III < IV$

B. $I < III < IV < II$

C. $I < IV < II < III$

D. $IV < I < II < III$

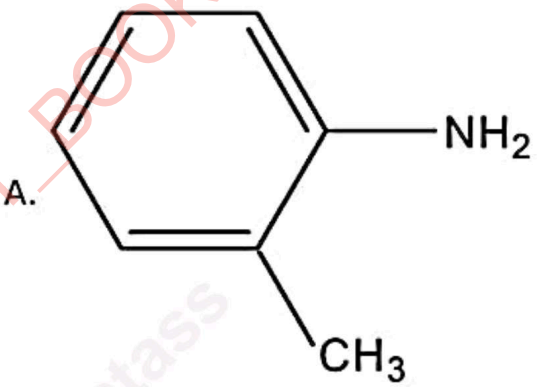
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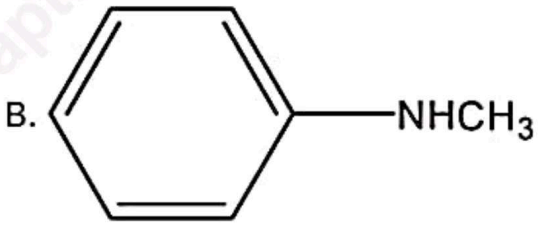
Question No. 13

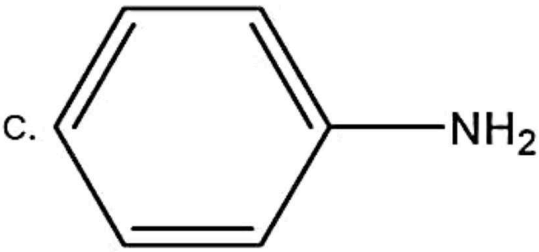
Only One Option Correct Type

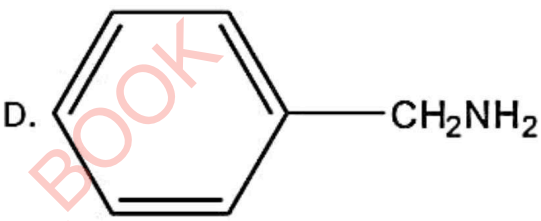
Each question has multiple options out of which ONLY ONE is correct.

Which of the following is the strongest base?

A. 

B. 

C. 

D. 

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Predict the compound C in the following series of reactions.

$$CH_3COOH \xrightarrow{PCl_5} [A] \xrightarrow[\text{Anhydrous } AlCl_3]{C_6H_6} [B] \xrightarrow{CH_3MgBr} [C]$$

A. $CH_3COC_6H_5$

B. $CH_3CH(OH)C_6H_5$

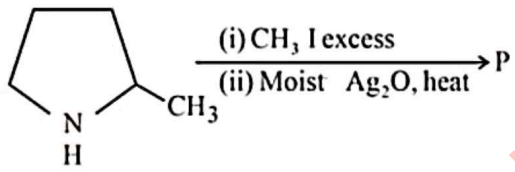
C. $CH_3C(CH_3)(OH)C_6H_5$

D. $CH_3CH(OH)C_2H_5$

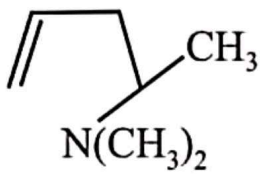
Question No. 15

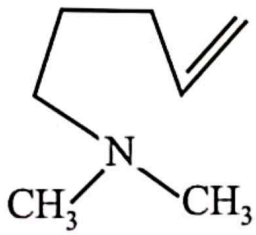
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

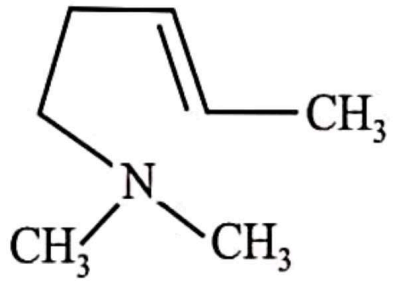


In the above reaction, the product P is

A. 

B. 

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C. 

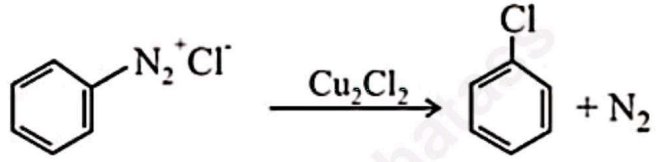
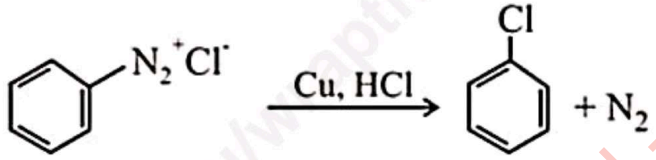
D. None of these

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match List-I with List-II.

List-I	List-II
(a) 	(i) Wurtz reaction
(b) 	(ii) Sandmeyer reaction
(c) $2CH_3CH_2Cl + 2Na \xrightarrow{Ether} C_2H_5 - C_2H_5 + 2NaCl$	(iii) Fittig reaction
(d) $2C_6H_5Cl + 2Na \xrightarrow{Ether} C_6H_5 - C_6H_5 + 2NaCl$	(iv) Gatterman reaction

Choose the correct answer from the options given below:

A. (a) → (iii), (b) → (iv), (c) → (i), (d) → (ii)

B. (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

C. (a) → (iii), (b) → (i), (c) → (iv), (d) → (ii)

D. (a) → (ii), (b) → (iv), (c) → (i), (d) → (iii)

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Pick out incorrect statement

A. In a electrolysis experiment, α -amino acids migrate at the isoelectric point towards electrodes

B. p-aminobenzenesulphonic acid as a dipolar ion; while p-aminobenzoic acid does not

C. Sulphanilic acid is soluble in base, but not in acid

D. H_3NCH_2COOH is more acidic than RCH_2COOH

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Among the following processes, find out the compound A, B, C and D, respectively.

(i) $CH_3CH_2Cl \xrightarrow{NaCN} A \xrightarrow[Ni/H_2]{\text{reduction}} B$

(ii) $C_6H_5NH_2 \xrightarrow{NaNO_2/HCl} C \xrightarrow{C_6H_5NH_2/OH^-} D$

A. $A = CH_3CN$, $B = CH_3CH_2NH_2$, $C = C_6H_5N_2^+Cl^-$, $D = C_6H_5N = NC_6H_4NH_2$

B. $A = CH_3CN,$
$B = CH_3CH_2CH_2NH_2,$
$C = C_6H_5N_2^+Cl^-,$
$D = C_6H_5N = NC_6H_4NH_2$

C. $A = CH_3CH_2CN,$
$B = CH_3CH_2CH_2NH_2,$
$C = C_6H_5N_2^+Cl^-,$
$D = C_6H_5N = NC_6H_4NH_2$

D. None of these.

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Ammonolysis of Alkyl halides followed by the treatment with NaOH solution can be used to prepare primary, secondary and tertiary amines. The purpose of NaOH in the reaction is.

A. to remove basic impurities

B. to activate NH_3 used in the reaction

C. to remove acidic impurities

D. to increase the reactivity of alkyl halide.

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the acidity of the carboxylic acids.
(I) $PhCOOH$
(ii) $o-NO_2C_6H_4COOH$
(iii) $p-NO_2C_6H_4COOH$
(iv) $m-NO_2C_6H_4COOH$
Which of the following order is correct?

A. $i > ii > iii > iv$

B. $ii > iv > iii > i$

C. $ii > iv > i > iii$

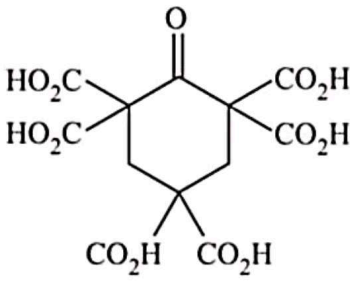
D. $ii > iii > iv > i$

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.



$\xrightarrow[\Delta]{}$ (n) moles of CO_2 evolved

during given reaction?
What is the value of 'n' here

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

A compound with molecular mass 180 is acylated with CH_3COCl to get a compound with molecular mass 390. The number of amino groups present per molecule of the former compound is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

How many of the following reagents react with arene diazonium salts to form aryl halides?

i. HBF_4/Δ

ii. KHS

iii. CuCN/KCN

iv. CuBr/HBr

v. Cu/HCl

vi. KI/Δ

vii. CH_3CH_2OH

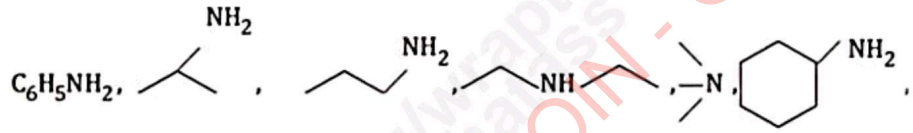
viii . H_3PO_2/H_2O

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

How many of these compounds can show positive test for Hinsberg reagent



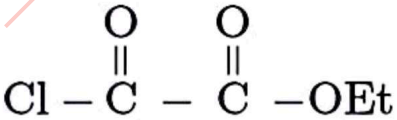
$(Ph)_3N, Ph-CH_2OH$

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Find the number of RMgX consumed/molecule with following reactant ?



Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\lim_{x\to 0} \frac{\left(\int_0^{x^2} e^{t^2-x}dt\right)-x^2e^{-x}}{\int_0^x \sin 2t(1-\cos 2t)^2 dt}$ is equal to -

A. 0

B. $\frac{1}{4}$

C. $\frac{1}{8}$

D. $\frac{1}{16}$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\int \frac{\sec^2 x dx}{(\tan^{101} x+\tan x)} = g(x)+c$ where $g\left(\frac{\pi}{4}\right) = -\frac{\ln 2}{100}$, then $\lim_{x\to \pi/2} g(x)$ is equal to

A. 0

B. 1

C. -1

D. 2

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f(x)$ is an even function which is also periodic with the period T and $\int_0^a f(x)\mathrm{dx} = 3$ and $\int_{-T/2}^{3T/2} f(x)\mathrm{dx} = 18$, then $\int_{-a}^{a+5T} f(x)\mathrm{d}x$ is equal to -

A. 96

B. 93

C. 51

D. 48

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $l_1 = \int_0^\infty \frac{x^2\sqrt{x}}{(1+x)^6}\,dx, l_2 = \int_0^\infty \frac{x\sqrt{x}}{(1+x)^6}\,dx$, then :

A. $l_1 = 2l_2$

B. $l_2 = 2l_1$

C. $l_1 = l_2^2$

D. $l_1 = l_2$

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\int \frac{x}{x+1+e^x}\,dx = px + q\ln|x+1+e^x| + c$, where c is the constant of integration, then $p+q$ is equal to

A. 0

B. 1

C. 2

D. 3

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $\int_0^9 e^x\left(1+\{x\}+\{x\}^2\right)dx = I$ and $\int_0^1 e^x\left(1+\{x\}+\{x\}^2\right)dx = J$, if $\frac{I}{J} = \sum_{n=0}^{k} e^n$, then k is equal to (where {.} denotes fractional part function)

A. 7

B. 8

C. 6

D. 9

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $\int_0^{2\pi} \min\{|x-\pi|, \cos^{-1}(\cos x)\}dx$ is-

A. $\frac{\pi^2}{4}$

B. $\frac{\pi^2}{2}$

C. $\frac{\pi^2}{8}$

D. π^2

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\int_0^{\pi} \sqrt{\left(\cos x + \cos 2x + \cos 3x\right)^2 + (\sin x + \sin 2x + \sin 3x)^2}$ has the value equal to $\left(\frac{\pi}{k} + \sqrt{w}\right)$ where k and w are positive integers, find the value of $\left(k^2 + w^2\right)$.

A. 153

B. 150

C. 144

D. 145

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of the integral $\int e^{3\sin^{-1}x}\left(\frac{1}{\sqrt{1-x^2}}+e^{3\cos^{-1}x}\right)dx$ is equal to
(where, c is an arbitrary constant)

A. $\frac{e^{3\sqrt{\sin^{-1}x}}}{3}+xe^{\frac{3\pi}{2}}+c$

B. $e^{\sqrt{\sin^{-1}x}}+e^{\pi/2}+c$

C. $\frac{e^{3\sin^{-1}x}}{3}+xe^{\frac{3\pi}{2}}+c$

D. $e^{\frac{\pi}{2}}+e^{x\left(\frac{\pi}{2}\right)}+c$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\int_{-3}^{2}\{|x+1|+|x+2|+|x-1|\}dx$ is equal to

A. $\frac{31}{2}$

B. $\frac{35}{2}$

C. $\frac{47}{2}$

D. $\frac{39}{2}$

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\int \frac{1}{\left(x^{5/6}+5x^{7/6}\right)\cdot\sqrt{1+4x^{1/3}}}\,dx$ is equal to -

A. $-6\tan^{-1}\left(x^{-1/3}+4\right)^{1/2}+C$

B. $-6\cot^{-1}\left(x^{-1/3}+4\right)^{1/2}+C$

C. $6\tan^{-1}\left(x^{1/3}+4\right)^{1/2}+C$

D. $6\cot^{-1}\left(x^{1/3}+4\right)^{1/2}+C$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f:R\to R$ such that
$f(x+2y)=f(x)+f(2y)+4xy, x, y\in R$ and $f'(0)=0$
If $I_1=\int_0^1 f(x)dx, I_2=\int_{-1}^0 f(x)dx, I_3=\int_{\sqrt{1/2}}^2 f(x)dx$, then

A. $I_1=I_2>I_3$

B. $I_1>I_2>I_3$

C. $I_1=I_2<I_3$

D. $I_1<I_2<I_3$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$$\int \frac{1}{3\cos x - 4\sin x + 5}\, dx =$$

A. $\frac{2}{\sqrt{5}} \tan^{-1}\left(\frac{3\tan\frac{x}{2}+4}{\sqrt{5}}\right) + c$

B. $\frac{3}{4} \tan^{-1}\left(\frac{\tan\frac{x}{2}}{3}\right) + c$

C. $\frac{1}{2-\tan\frac{x}{2}} + c$

D. $\frac{1}{1+\tan\frac{x}{2}} + c$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of the integral $\int \frac{x^2 - 4x\sqrt{x} + 6x - 4\sqrt{x} + 1}{x - 2\sqrt{x} + 1}\, dx$ is equal to

A. $\frac{x^{\frac{3}{2}}}{2} + x + c$

B. $\frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}} + x + c$

C. $x^{\frac{3}{2}} + \frac{x}{2} + c$

D. $\frac{2}{3}x^{\frac{3}{2}} + c$

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Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the function

$f(x) = \int_0^x \left(5\,In\left(1+t^2\right) - 10t\tan^{-1}t + 16\sin t\right)dt.$ $f(x)$ is

A. Negative for all $x \in (0,1)$

B. Increasing for all $x \in (0,1)$

C. Non-monotonic for all $x \in (0,1)$

D. None of these

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If

$\int \frac{x-\sin x}{1+\cos x}\,dx = x\tan\left(\frac{x}{2}\right) + p\log\left|\sec\left(\frac{x}{2}\right)\right| + C,$

then p is equal to

A. -4

B. 4

C. 2

D. -2

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The integral $I = \int_0^{100\pi} \left[\tan^{-1} x\right] dx$, (where, [.] represents the greatest integer function) has the value $K\pi + \tan p$, then the value of $K + p$ is equal to

A. 101

B. 99

C. 100π

D. 99π

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\mathrm{I_1} = \int_0^{\pi} \frac{\mathrm{x\ \sin\ x}}{1+\cos^2 \mathrm{x}}\,\mathrm{dx},\ \mathrm{I_2} = \int_0^{\pi} \mathrm{x\ \sin^4 x\ dx}$ then $\mathrm{I_1 : I_2} =$

A. $3 : 4$

B. $1 : 2$

C. $4 : 3$

D. $2 : 3$

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u/wrotthatass

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} \cdot \frac{1}{x}\, dx = A \ln\left|\frac{1+\sqrt{1-x}}{\sqrt{x}}\right| + B\cos^{-1}\sqrt{x} + C$, then value of $A + B$ is-
(where C is an integertion constant)

A. 2

B. -2

C. 4

D. 0

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be a differentiable non- decreasing function such that $\int_0^x (f(t))^3 dt = \frac{1}{x^2}\left(\int_0^x f(t)dt\right)^3 \forall x \in R - \{0\}$ and $f(1) = 1$. If $\int_0^x f(t)dt = g(x)$ then $\frac{xg'(x)}{g(x)}$ is

A. Always equal to 1

B. Always equal to -2

C. May be 1 or -2

D. Not independent of x

Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

If the integral $\int \frac{5\tan x}{\tan x - 2}\,dx = x + a\ln\left|\sin x - 2\cos x\right| + k$, then a is equal to :

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Let f be a continuous and differentiable function in (x_1, x_2). If $f(x).f'(x) \geq x\sqrt{1-(f(x))^4}$ and $\lim\limits_{x\to x_1^+}(f(x))^2 = 1$ and $\lim\limits_{x\to x_2^-}(f(x))^2 = \frac{1}{2}$ then minimum value of $x_1^2 - x_2^2$ is λ then $\frac{3\lambda}{\pi}$ equals to .....

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Let $F(x) = \int\limits_{x}^{\frac{\pi}{3}+x^2} 2\cos^2 t\,dt$, $x \in \mathrm{R}$ and $f : \left[0, \frac{1}{2}\right] \to \left[0, \infty\right)$ be a continuous function. For $a \in \left[0, \frac{1}{2}\right]$. If $\mathrm{F}'(a) + 2$ is the area of the region bounded by $y = f(x)$, $x = 0$, x-axis and $x = a$, then find $f(0)$.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

The value of the definite integral, $\int_{\theta_1}^{\theta_2} \frac{\mathrm{d}\theta}{1+\tan\theta} = \frac{501\pi}{\mathrm{K}}$ where $\theta_2 = \frac{1003\pi}{2008}$ and $\theta_1 = \frac{\pi}{2008}$. The value of K equals

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

$\int\left(\mathrm{x}^{12}+\mathrm{x}^{8}+\mathrm{x}^{4}\right)\left(2\mathrm{x}^{8}+3\mathrm{x}^{4}+6\right)^{1/4}\mathrm{dx} = \mathrm{a}\left(2\mathrm{x}^{12}+3\mathrm{x}^{8}+6\mathrm{x}^{4}\right)^{5/4}$ (where C is integration constant), then the value of $(60a+1)$ is

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uestion No. 4

Numerical Type

The answer has to be filled into the input box provided below.

The value of the definite integral, $\int_{\theta_1}^{\theta_2} \frac{d\theta}{1+\tan\theta} = \frac{501\pi}{K}$ where $\theta_2 = \frac{1003\pi}{2008}$ and $\theta_1 = \frac{\pi}{2008}$. The value of K equals

uestion No. 5

Numerical Type

The answer has to be filled into the input box provided below.

$\int \left(x^{12}+x^8+x^4\right)\left(2x^8+3x^4+6\right)^{1/4} dx = a\left(2x^{12}+3x^8+6x^4\right)^{5/4} + C$ where C is integration constant), then the value of $(60a+1)$ is

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